

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 2	Team No- 32	Project Title- Online Rental Contract Booking & Payment System for Rental & Leasing
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1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The **Online Rental Contract Booking & Payment System for Rental & Leasing** is an intelligent web-based platform designed to digitize and simplify the complete rental lifecycle for tenants, property owners, and administrators. The system enables users to search and filter properties based on location, budget, property type, amenities, and availability, view detailed property information, book properties online, generate digital rental contracts, and make secure online payments.

A key feature is the **AI-powered rental assistant**, which understands natural-language queries and recommends suitable properties based on user preferences, budget, and requirements. The system also provides **nearby rental price comparison**, allowing users to compare rental amounts, distances, amenities, and property types of similar properties in the surrounding area. Through **Google Maps and Navigation integration**, users can visualize properties on an interactive map, calculate distances, and obtain navigation directions to selected properties.

The platform provides property owners with tools to manage listings, bookings, contracts, payments, and availability, while administrators can monitor users, properties, transactions, and system activities. Secure authentication, role-based access control, transaction management, and payment tracking ensure data security and reliability.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030572	S.Harshith	
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3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- ☐ **Frontend:** React.js, HTML5, CSS3, JavaScript, Bootstrap/Tailwind CSS
- ☐ **Backend:** FastAPI (Python), Node.js with Express.js, Spring Boot
- ☐ **Databases:** PostgreSQL, MongoDB
- ☐ **Vector Database:** pgvector Pinecone for AI-powered semantic search and recommendations
- ☐ **AI/ML:** OpenAI API Python AI libraries for rental assistance and property recommendations
- ☐ **API & Integration:** REST API, Google Maps Platform, Google Navigation/Routes API
- ☐ **Authentication & Security:** JWT, OAuth 2.0, Role-Based Access Control (RBAC), password hashing
- ☐ **ORM/Database Tools:** SQLAlchemy, Mongoose, JPA/Hibernate
- ☐ **Payment Gateway:** Razorpay / Stripe
- ☐ **Microservices & Communication:** FastAPI, Node.js, Spring Boot, REST-based service communication
- ☐ **Containerization:** Docker and Docker Compose

- ☐ **Testing:** Pytest, Jest, Postman
- ☐ **API Documentation:** OpenAPI / Swagger
- ☐ **Monitoring:** Prometheus, Grafana, OpenTelemetry
- ☐ **CI/CD:** GitHub Actions
- ☐ **Version Control:** Git and GitHub
- ☐ **Development Tools:** Visual Studio Code, IntelliJ IDEA, Postman
- ☐ **Deployment Platform:** Docker-compatible cloud platform such as AWS / Azure / Google Cloud.

Guide Name and Signature (remarks if any) _____